Name: Ryan O'Connell

Q1. Scarlet Bakery bakes bagels and muffins. A dozen bagels uses 2 kg of flour and 1 hour of oven time; a dozen muffins uses 1 kg of flour and 1 hour of oven time. Each day 40 kg of flour and 30 oven hours are available, and at least 5 dozen muffins must be baked. Profit is $9 per dozen bagels and $7 per dozen muffins. Formulate an LP that maximizes daily profit.

(a) Define the decision variables. [8 pts]

x1 = dozens of bagels per day, x2 = dozens of muffins per day

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(b) Write the objective function. [8 pts]

max z = 9x1 + 7x2

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(c) Write all constraints. [14 pts]

2x1 + x2 <= 40; x1 + x2 <= 30; x2 <= 5; x1, x2 >= 0

"At least 5 dozen muffins" is x2 >= 5, not <=.

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Q2. Convert to standard form: max z = x1 + 4x2 subject to -x1 + x2 <= 3, 3x1 + x2 >= 6, x1 >= 0, x2 free.

(a) Write the constraints as equalities with slack/surplus variables. [10 pts]

-x1 + x2 + s1 = 3; 3x1 + x2 + s2 = 6

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Sign of the slack variable is reversed: a >= constraint needs a surplus variable subtracted (-s), not a slack added (+s).

(b) Handle the free variable x2 and rewrite the objective. [10 pts]

x2 >= 0; max z = x1 + 4x2

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x2 is free: substitute x2+ - x2- instead of assuming x2 >= 0.

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Q3. Simplex: max z = 2x1 + 3x2 subject to x1 + 2x2 <= 8, 3x1 + 2x2 <= 12, x1, x2 >= 0.

(a) Choose the entering and leaving variables for the first pivot. [10 pts]

x1 enters; s2 leaves.

The most negative z-row coefficient is -3 (x2), so x2 enters.

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(b) Carry out pivots to optimality. [10 pts]

Optimal x1 = 2, x2 = 4.

Check the second pivot: after updating, the x2 row is (0, 1, 3/4, -1/4 | 3).

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(c) State the optimal solution and value; explain how you know it is optimal. [10 pts]

x1 = 0, x2 = 4, z = 12 - optimal after one pivot.

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Q4. Graphical method: max z = 5x1 + 5x2 subject to x1 + x2 <= 8, x1 <= 5, x2 <= 6, x1, x2 >= 0.

(a) List the corner points of the feasible region. [10 pts]

(0, 0), (5, 0), (5, 4), (2, 6), (0, 6)

x1 = 5 meets x1 + x2 = 8 at x2 = 3, not 4.

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(b) Find all optimal solutions and explain. [10 pts]

Every point on the segment between (2, 6) and (5, 3) is optimal with z = 40: the objective is parallel to x1 + x2 <= 8.

10/10